

UNIT-1 MINI PROJECT

Enterprise Student Record and Academic Management Platform

Design Thinking, Product Planning & DevOps-Ready Engineering

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Subject	DevOps – Unit 1 Mini Project

Prepared according to the Unit-1 Mini Project Guide

1. Project Overview

Project Title: Enterprise Student Record and Academic Management Platform (ESRAMP)

This document presents the Unit-1 mini project for the DevOps & Cloud subject. The assignment asks students to move through the early, pre-coding stages of building a software product — understanding the business domain, identifying stakeholders, applying Design Thinking, defining requirements, and preparing a product vision, MVP and roadmap — for an Enterprise Student Record and Academic Management Platform. The reference training guide illustrates this same methodology using a Library Management Platform; that example is used only to understand the expected depth and structure, and is not the subject of this report.

Scope note: This is a Unit-1 product-discovery and planning exercise. The architecture, repository layout and roadmap presented below are proposed engineering plans for later units, not a claim that the platform has already been built or deployed.

1.1 Project Objectives

- Design a single, centralized platform for student academic records.
- Reduce reliance on manual registers and disconnected spreadsheets.
- Support student registration, attendance, examinations, results and academic reporting.
- Provide role-based access for students, faculty, academic administrators, examination staff and IT support.
- Lay a clear product foundation that later units can carry into Agile development and DevOps automation.

1.2 Business Problem Statement

Higher-education institutions handle a large volume of student information across admissions, course registration, attendance, examinations, results and reporting. When this information lives in manual registers or disconnected files, it becomes easy to duplicate, hard to keep consistent, and slow to retrieve — and students and faculty have no single reliable view of academic data.

Core Problem Statement: Students, faculty and academic administrators need a centralized, reliable way to manage academic information, because fragmented and manual record-keeping causes delays, duplicate entries, reporting overhead and poor visibility for everyone involved.

2. Business Requirements Document (BRD)

2.1 Business Understanding

The platform is scoped around the academic process areas named in the project guide: student admission, course registration, attendance, examinations, result management and academic reporting.

Business Area	Current / Expected Activity	Platform Need
Student Admission & Registration	Create and maintain each student's academic profile.	Central student profile and registration module.
Course Registration	Assign students to programs, courses and subjects.	Course and enrollment management.
Attendance	Record attendance against subjects and students.	Faculty attendance entry and student-facing view.
Examination	Track examination-related academic information.	Examination and assessment records.
Result Management	Store and publish marks and results.	Result entry, calculation and publishing workflow.
Academic Reporting	Prepare student and administrative	Searchable, export-ready report

Business Area	Current / Expected Activity	Platform Need
	reports.	generation.

2.2 Business Objectives

- Centralize academic records in one platform.
- Make authorized student information easier to reach for the people who need it.
- Cut down repeated manual entry and reporting effort.
- Improve consistency and traceability of academic records.
- Build a foundation that can later adopt CI/CD, containers, infrastructure automation and monitoring.

2.3 Expected Business Benefits

- Faster retrieval of student academic information.
- Better visibility for faculty and academic administrators.
- Fewer duplicate entries and avoidable data-entry mistakes.
- More structured, repeatable academic reporting.
- Clear ownership and access boundaries across roles.
- A development path that can grow from product discovery into Agile delivery and DevOps practice.

3. Stakeholder Analysis Report

The project guide names students, faculty, academic administrators, examination cell staff and IT support as the core stakeholder groups. Their expected involvement is summarized below.

Stakeholder	Role	Needs / Expectations	Project Responsibility
Students	Primary users	View profile, attendance, results and academic reports.	Use the platform and provide usability feedback.
Faculty Members	Academic users	Manage attendance and academic data for assigned subjects.	Enter and verify academic data, validate workflows.
Academic Administrators	Management users	Maintain student/course records, generate reports.	Approve processes and monitor data quality.
Examination Cell Staff	Examination operations	Manage examination and result information.	Maintain examination records and result workflows.
IT Support	Operations	A reliable, secure, maintainable platform.	User support, deployment and operational upkeep.
College Management	Business stakeholder	Visibility, compliance and improved academic operations.	Set priorities and evaluate outcomes.

4. User Persona Report

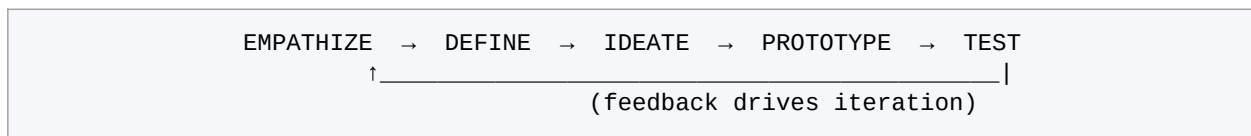
These are representative, conceptual personas built from the stakeholder analysis above — not records of real interviews — used to turn stakeholder needs into concrete, user-centred requirements.

Persona	Goals	Responsibilities	Challenges	Expected Features
Rehan — Student	Access academic information quickly and reliably.	View personal academic records and reports.	Hunting for information spread across multiple sources.	Student dashboard, attendance, results, reports.

Persona	Goals	Responsibilities	Challenges	Expected Features
Sneha — Faculty Member	Keep accurate academic records for her subjects.	Record attendance and academic information.	Repeated manual entry and re-verification.	Faculty dashboard, attendance entry, subject-wise records.
Dr. Kulkarni — Academic Administrator	Monitor academic operations across the institution.	Manage records and generate reports.	Manual report preparation, limited centralized visibility.	Admin dashboard, search, reports and controls.
Mr. Joshi — Examination Staff	Manage examination and result data accurately.	Enter and verify examination information.	Data consistency and result-processing workload.	Exam/result module, validation and report generation.

5. Design Thinking Report

Following the guide's methodology, the platform is planned through five stages — Empathize, Define, Ideate, Prototype and Test — before any large-scale coding begins.



5.1 Empathize

The first step is understanding the academic environment as experienced by students, faculty, administrators, examination staff and IT support — where information is created, how it is updated, who needs it, and where delays or confusion occur.

- Observe how students currently try to access academic information.
- Understand faculty attendance and record-keeping workflows.
- Understand administrative and examination reporting needs.
- Identify the friction caused by manual or fragmented record management.

5.2 Define

Defined problem: the institution needs a centralized academic-record platform, because fragmented and manual handling of student, attendance, examination and result data increases effort, delays access, and makes reporting difficult.

How Might We questions:

- How might we give students a single place to view their academic information?
- How might we let faculty record attendance with less repeated manual effort?
- How might we make examination and result data consistent and easy to verify?
- How might we help administrators generate academic reports without manual compilation?

5.3 Ideate

- Central student profile and academic record.
- Role-based dashboards for students, faculty and administrators.
- Attendance management with subject-wise views.
- Examination and result management.
- Academic report generation.
- Search and filtering for authorized users.
- Notifications for academic updates (future enhancement).

5.4 Prototype

A low-fidelity prototype (conceptual, not yet built) is proposed to cover the following screens before implementation:

- Login / role selection screen.
- Student dashboard.
- Faculty attendance dashboard.
- Administrator dashboard.
- Student record / search screen.
- Examination and result screen.
- Academic report screen.

5.5 Test

Proposed User Research / Design Exercise: the prototype should be walked through with representative users, focusing on navigation clarity, ease of finding records, attendance entry, result viewing, and report generation. Feedback would be logged and fed into the next iteration. No usability testing has actually been conducted at this stage.

6. IBM Design Thinking Report

The guide requires the three IBM Design Thinking principles: Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams.

6.1 Focus on User Outcomes

Success is measured by user outcomes rather than technical completion alone.

Feature-Centric Framing	User-Outcome Framing
Build a student database search.	Let an authorized user find a student's academic information quickly, in a few clicks.
Create an attendance entry form.	Let faculty confirm a class's attendance in under a minute, without duplicate entry.
Add a result publishing module.	Let students see verified results as soon as they are approved, without asking staff.

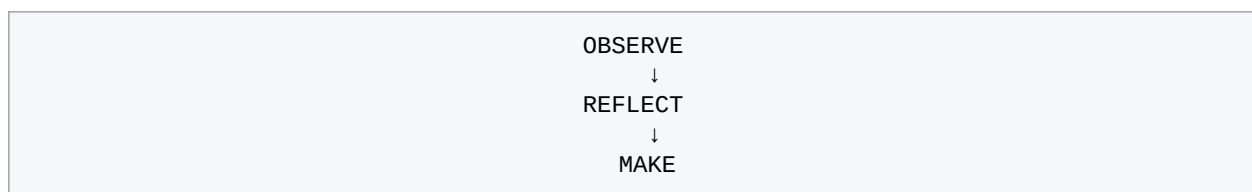
6.2 Restless Reinvention

The platform is treated as an evolving product, not a one-time build. After each development or release cycle, user feedback, usage patterns, defects and performance data would be reviewed, and improvements planned for the next iteration — continuous UI refinement, backend improvement and incremental automation of deployment.

6.3 Diverse Empowered Teams

A cross-functional team spanning product/academic representatives, UX contributors, software engineers, QA, database administrators, DevOps engineers and security reduces communication gaps and keeps decisions grounded in both user needs and engineering constraints.

6.4 IBM Loop Documentation — Observe → Reflect → Make



(back to OBSERVE)	
Loop Stage	Application to ESRAMP
Observe	Study user interactions, academic workflows, feedback, logs and operational problems.
Reflect	Analyze feedback and metrics, identify priorities, and compare outcomes against requirements.
Make	Design or modify the product, implement changes, test them, and prepare the next release.

7. Functional Requirements Specification (FRS)

Functional requirements are drawn from the guide's examples of registration, attendance, examinations, results and reporting, organized into a practical MVP set.

ID	Functional Requirement	Description	Priority
FR-01	User Authentication	Authorized users can sign in according to their assigned role.	High
FR-02	Student Registration	Create and maintain student registration records.	High
FR-03	Student Profile	Store and display authorized student academic information.	High
FR-04	Course Registration	Maintain student course/subject enrollment data.	High
FR-05	Attendance Management	Faculty can record and view attendance for assigned subjects.	High
FR-06	Examination Management	Maintain examination and assessment information.	Medium
FR-07	Result Management	Enter, verify and display student results.	High
FR-08	Report Generation	Generate student and administrative academic reports.	High
FR-09	Search & Filter	Authorized users can search relevant records efficiently.	Medium
FR-10	Role-Based Access	Limit features and records according to user role.	High

8. Non-Functional Requirements Specification (NFRS)

ID	Requirement	Proposed Target / Expectation
NFR-01	Security	Role-based access, secure authentication, protection for sensitive academic data.
NFR-02	Usability	Simple navigation and clear, role-specific dashboards.
NFR-03	Reliability	Consistent record storage and validation with backup/recovery planning.
NFR-04	Performance	Common searches and dashboard actions should respond quickly under expected load.
NFR-05	Scalability	Architecture should absorb more users, courses and records without a redesign.
NFR-06	Maintainability	Modular code, documented repository structure, tests, clear separation of concerns.

ID	Requirement	Proposed Target / Expectation
NFR-07	Availability	Deployment designed for dependable access during normal academic operations.
NFR-08	Auditability	Important academic changes traceable through logs/history.

9. Product Vision Document

9.1 Vision Statement

To give every student, faculty member and academic administrator a centralized, secure and easy-to-use platform for managing student records, attendance, examinations, results and academic reports across the institution.

9.2 Mission Statement

To simplify academic record management by connecting students, faculty and administrators through reliable digital workflows, while building an engineering foundation that can later be developed and operated using Agile and DevOps practices.

9.3 Business Objectives

- Centralize academic information.
- Improve accessibility of authorized student records.
- Reduce manual effort in attendance and reporting.
- Improve consistency and traceability of academic data.
- Provide a foundation for iterative development and automated delivery.

9.4 Success Criteria (Proposed Project Targets)

- Users can complete core MVP workflows without unnecessary manual steps.
- Authorized users can locate required academic records efficiently.
- Attendance and result records are stored consistently.
- Academic reports can be generated from centralized data.
- The repository and project process are organized for future Agile/DevOps work.

10. Minimum Viable Product (MVP) Definition

The MVP is the first practical release, limited to the features needed to demonstrate the platform's core value.

MVP Feature	Included Capability	Primary Users
User Login & Roles	Authentication and role-based access.	All users
Student Registration	Create/update student academic profile.	Admin / authorized staff
Record Management	Store and view core student records.	Admin / faculty / student
Attendance	Record and view attendance.	Faculty / student
Examination & Results	Manage and display result information.	Exam staff / faculty / student
Academic Reports	Generate basic academic reports.	Admin / student / authorized staff

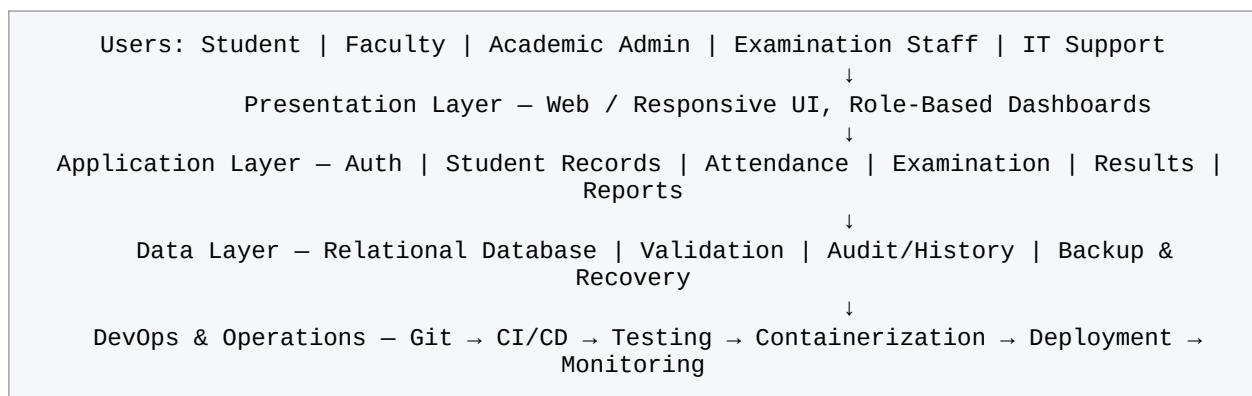
Deferred to later releases: advanced analytics, AI-based recommendations, complex third-party integrations and other non-essential capabilities — kept out of the first MVP so the core product can be validated first.

11. Product Roadmap

Phase	Focus	Major Deliverables
Phase 1	Domain & Product Discovery	Business understanding, stakeholders, personas, problem statement, BRD.
Phase 2	Design Thinking & Prototyping	Empathy findings, ideation, wireframes, prototype feedback.
Phase 3	Agile Planning	Backlog, user stories, sprint planning, acceptance criteria.
Phase 4	Core Development	Frontend, backend, database and MVP workflows.
Phase 5	DevOps Automation	Git workflow, CI pipeline, automated tests, containerization, CD planning.
Phase 6	Infrastructure Automation	Infrastructure-as-Code and configuration management.
Phase 7	Deployment & Monitoring	Production deployment, logging, monitoring, continuous improvement.

12. Initial System Architecture Diagram

Proposed architecture for later implementation — intentionally technology-neutral at Unit-1 level, showing the major engineering layers only. Labelled: Initial Conceptual System Architecture.



Data flow: users access role-specific dashboards → application services validate permissions and process academic operations → the database stores authorized records → reports and dashboards present the required information → DevOps tooling supports testing, delivery, deployment and monitoring.

13. Project Charter

Project Name	Enterprise Student Record and Academic Management Platform
Project Owner / Student	Akash Suresh Dhanwate
PRN	2125UDSM1024
Purpose	Design a centralized platform for managing student academic information.
Primary Scope	Student registration, records, attendance, examinations, results and reports.
Primary Users	Students, faculty, academic administrators, examination staff and IT support.
Success Direction	Usable MVP, clear requirements, organized repository and a DevOps-ready engineering plan.
Constraints	Unit-1 emphasizes discovery and product planning; full implementation is planned for subsequent units.
Key Risks	Incomplete requirements, access-control mistakes, inconsistent data, scope expansion, insufficient user feedback.

Risk Response	Iterative requirements review, role-based design, validation, testing, user feedback, controlled MVP scope.
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13.1 Milestone Plan

Milestone	Outcome
M1	Business requirements and problem statement completed.
M2	Stakeholders and personas documented.
M3	Design Thinking and IBM Design Thinking documentation completed.
M4	FRS/NFRS and product vision completed.
M5	MVP and roadmap finalized.
M6	Initial architecture and repository structure prepared.
M7	Project ready for Agile development and later DevOps implementation.

14. Project Repository Initialization

A standardized repository makes the project easier to understand, build, test and operate. The following structure is proposed for this mini project and later development phases.

```
student-record-academic-platform/
|-- .github/workflows/      (future CI/CD workflows)
|-- app/
|   |-- frontend/          (user interface)
|   |-- backend/           (application / API services)
|   |-- modules/           (students, attendance, exams, results, reports)
|   |-- tests/             (unit and integration tests)
|-- database/
|   |-- schema/            (database design)
|   |-- migrations/        (future schema migrations)
|-- docs/                  (BRD, FRS, NFRS, personas, architecture)
|-- docker/                (future containerization files)
|-- infrastructure/         (future IaC files)
|-- monitoring/            (future monitoring configuration)
|-- README.md
`-- .gitignore
```

14.1 Git Initialization Plan

- Create the project repository with the standardized folder structure.
- Add a README.md containing project objective, scope and setup information.
- Create the initial commit for the Unit-1 documentation.
- Use meaningful branches and commits once development begins.
- Keep application code, documentation and infrastructure configuration in separate folders.

15. Product Discovery Documentation

15.1 Discovery Questions

- Who creates and updates student academic records?
- What information does a student need most often?
- How is attendance currently recorded and verified?
- How are examination and result records prepared?
- Which reports do students, faculty and administrators actually need?

- Which roles should view or modify each type of record?
- What are the most common causes of delay or duplicate data?
- What feedback should be gathered before the prototype is finalized?

15.2 Assumptions to Validate

- The institution requires role-based access to academic information.
- Student, faculty and administrator workflows are meaningfully different.
- Core MVP records can be represented in a structured relational data model.
- User feedback will be available during prototype testing.
- Later units will define detailed technology and deployment choices.

16. Deliverable Traceability Matrix

Guide Deliverable	Where Covered in This Report
Business Requirements Document (BRD)	Section 2
Product Vision Document	Section 9
Stakeholder Analysis Report	Section 3
User Persona Report	Section 4
Functional Requirements Specification (FRS)	Section 7
Non-Functional Requirements Specification (NFRS)	Section 8
Design Thinking Report	Section 5
IBM Design Thinking Report	Section 6
IBM Loop Documentation	Section 6.4
MVP Definition	Section 10
Product Roadmap	Section 11
Initial System Architecture Diagram	Section 12
Project Charter	Section 13
Project Repository Initialization	Section 14
Project Folder Structure	Section 14
Product Discovery Documentation	Section 15

17. Conclusion

The Enterprise Student Record and Academic Management Platform has been planned here as a product-engineering exercise, not just a coding task. The report starts from the academic domain and business problem, identifies stakeholders and personas, works through the five Design Thinking stages, documents IBM Design Thinking and the Observe → Reflect → Make loop, defines functional and non-functional requirements, sets out a product vision and MVP, and closes with a phased roadmap and repository structure.

The result is a structured, DevOps-ready foundation for the stages ahead. Actual implementation, the CI/CD pipeline, containerization, infrastructure automation, deployment and monitoring are left for later units, in line with the course's progression.

18. Source Basis

This report is built primarily from the Unit-1 Mini Project brief, which specifies the objective, activities and required deliverables for the Enterprise Student Record and Academic Management Platform. The content above follows that structure and fills in the areas the brief leaves open for the student to design.

The DevOps Unit-1 Mini Project training/guide document was used only as a methodology and depth reference — it works through the same lifecycle using a Library Management Platform as its illustrative example, and that example does not appear as the subject of this report.